

ECO102: Introduction to Macroeconomics

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SECTION: 05

LECTURE 09

CHAPTER 27: AGGREGATE SUPPLY

Aggregate Supply

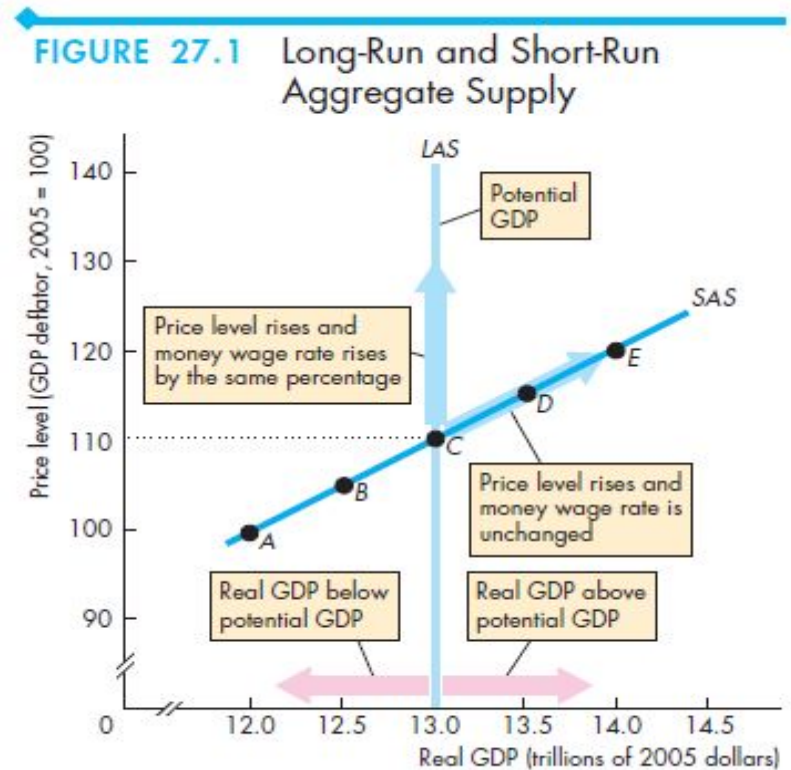
- ▶ The *quantity of real GDP supplied* is the total quantity of goods and services, valued in constant base year (2005) dollars, that firms plan to produce during a given period.
- ▶ This quantity depends on the quantity of labor employed, the quantity of physical and human capital, and the state of technology.
- ▶ It depends on decisions made by households and firms about the supply of and demand for labor.
- ▶ The labor market can be in any one of three states: at full employment, above full employment, or below full employment.
- ▶ *Aggregate supply* is the relationship between the quantity of real GDP supplied and the price level.
- ▶ This relationship is different in the long run than in the short run. Therefore we distinguish between the two time frame:
 - Long-run aggregate supply
 - Short-run aggregate supply

Long Run Aggregate Supply

- ▶ **Long-run aggregate supply** is the relationship between the quantity of real GDP supplied and the price level when the money wage rate changes in step with the price level to maintain full employment.
- ▶ The quantity of real GDP supplied at full employment equals potential GDP and this quantity is the same regardless of the price level.
- ▶ The long-run aggregate supply curve is vertical because potential GDP is independent of the price level.
- ▶ the *LAS* curve is accompanied by a change in *two* sets of prices: the prices of goods and services—the price level—and the prices of the factors of production, most notably, the money wage rate.
- ▶ So, when the price level changes and the real wage rate remains constant, employment remains constant and real GDP remains constant at potential GDP.

Short-Run Aggregate Supply

- ▶ **Short-run aggregate supply** is the relationship between the quantity of real GDP supplied and the price level *when the money wage rate, the prices of other resources, and potential GDP remain constant*.
- ▶ In the short run, a rise in the price level brings an increase in the quantity of real GDP supplied. The short-run aggregate supply curve slopes upward.
- ▶ With a given money wage rate, there is one price level at which the real wage rate is at its full-employment equilibrium level.
- ▶ At this price level, the quantity of real GDP supplied equals potential GDP and the *SAS* curve intersects the *LAS* curve.

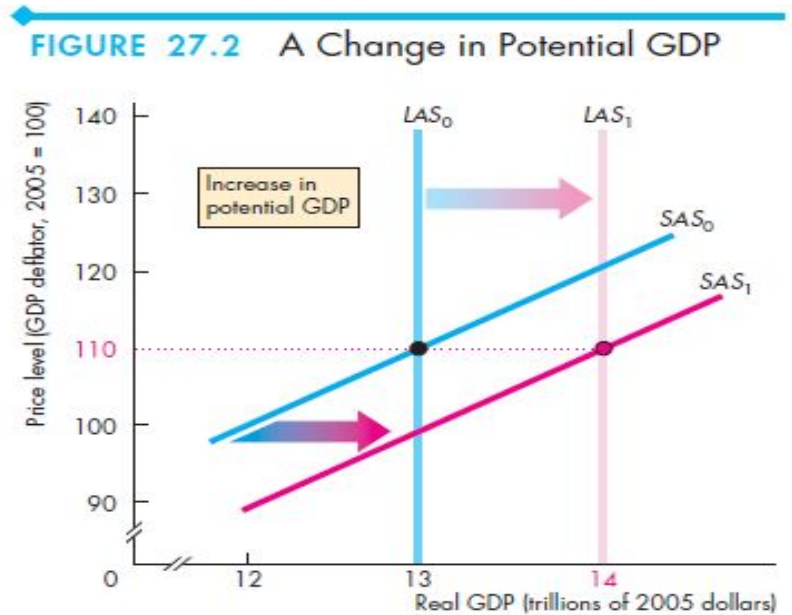


Changes in Aggregate Supply

- ▶ A change in the price level changes the quantity of real GDP supplied, which is illustrated by a movement along the short-run aggregate supply curve.
- ▶ It does not change aggregate supply.
- ▶ Aggregate supply changes when an influence on production plans other than the price level changes.
- ▶ These other influences include changes in potential GDP and changes in the money wage rate.

Changes in Potential GDP

- ▶ An increase in potential GDP increases both long-run aggregate supply and short-run aggregate supply.
- ▶ The two supply curves shift by the same amount only if the full-employment price level remains constant, which we will assume to be the case.
- ▶ Potential GDP can increase for any of three reasons:
 - An increase in the full-employment quantity of labor
 - An increase in the quantity of capital
 - An advance in technology

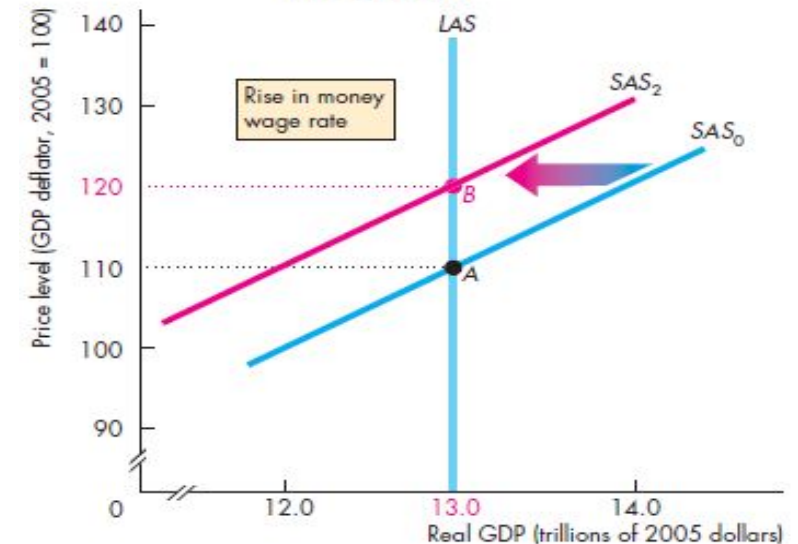


An increase in potential GDP increases both long-run aggregate supply and short-run aggregate supply. The long-run aggregate supply curve shifts rightward from LAS_0 to LAS_1 and the short-run aggregate supply curve shifts from SAS_0 to SAS_1 .

Changes in the Money Wage Rate

- ▶ When the money wage rate (or the money price of any other factor of production such as oil) changes, short-run aggregate supply changes but long-run aggregate supply does not change.
- ▶ A rise in the money wage rate decreases short-run aggregate supply because it increases firms' costs. With increased costs, the quantity that firms are willing to supply at each price level decreases, which is shown by a leftward shift of the SAS curve.
- ▶ A change in the money wage rate does not change long-run aggregate supply because on the LAS curve, the change in the money wage rate is accompanied by an equal percentage change in the price level.
- ▶ The money wage rate can change for two reasons: departures from full employment and expectations about inflation.

FIGURE 27.3 A Change in the Money Wage Rate



A rise in the money wage rate decreases short-run aggregate supply and shifts the short-run aggregate supply curve leftward from SAS_0 to SAS_2 . A rise in the money wage rate does not change potential GDP, so the long-run aggregate supply curve does not shift.